Thematic Analysis of Research Poster Motivation

Executive Summary

This report presents a thematic analysis of 27 student research posters from a genomic data science training program. The analysis categorized research posters based on their primary motivation into six categories: five predefined thematic categories and one 'Other' category for posters that did not clearly fit the main categories.

Category Distribution and Definitions

Category	Count	Definition
Disease-Motivated	['Avetisyan et al. (2023) - CCC SP23.txt', 'Brown et al. (2025) - CCC SP25.txt', 'Cabral et al. (2025) - CC	Research focused on understanding or treating specific human diseases or disor
Metabolic/Physiological	['Berbouti et al. (2025) - CCC FA25.txt', 'Deal et al. (2025) - LU SP25.txt', 'Lessing et al. (2025) - FA	Research focused on fundamental metabolic processes, nutrient processing, energy balance
Basic Biology	['Diaz et al. (2025) - CCC SP25.txt', 'Ford et al. (2023) - CCC SP23.txt', 'Sakamoto et al. (2025) - CCC SP2	Research focused on fundamental biological mechanisms, gene function, developmental
Other	0	No posters assigned to this category

Posters by Category with Supporting Evidence

Disease-Motivated (19 posters)

Avetisyan et al. (2023) - CCC SP23.txt

- Focus on autism-related genes and the gut-microbiome-brain axis
- Explicit connection to Autism Spectrum Disorder (ASD)
- Research linked to understanding autism treatment through gut intervention

Brown et al. (2025) - CCC SP25.txt

- Directly studies acid reflux disorder
- States goal is to 'lay the foundations for future studies that could lead to more understanding and treatment of acid reflux'
- Connects genetic findings to human gastric acid disorders

Cabral et al. (2025) - CCC FA25.txt

- Focus on MECP2 gene linked to Rett syndrome
- Explicitly studies neurodevelopmental disorders
- Discusses how 'dysregulation of this gene may contribute to neurological dysfunction'

Camara et al. (2025) - LU SP25.txt

- States amylase genes 'may relate to diseases like diabetes and Alzheimer's'
- Explicitly connects to human disease relevance

- Discusses gut-brain axis connection to neurodegenerative diseases

Gill & Alcazar (2025) - CCC FA25.txt

- Studies serotonin receptor genes linked to mental health disorders
- Explicitly references 'mental health disorders' and mood regulation
- Discusses prefrontal cortex function in emotional regulation

Godoy-Pena et al. (2025) - CCC FA25.txt

- Focuses on genes dictating sleep quality
- Explicitly connects to sleep disorders and their health impacts
- Studies 'poor diet, low sleep quality, and reduced energy levels'

Haubelt & Alcazar et al. (2025) - CCC FA25.txt

- Directly studies Parkinson's Disease and dopaminergic synapse
- States goal is investigating 'the specific genes involved in the dopaminergic synapse' related to Parkinson's
- Explicitly connects gene variants to neuropsychiatric disorders

Henriquez et al. (2025) - COD WI24.txt

- Explicitly mentions 'helping in understanding human immune disorders that reside within the gut such as Crohn's disease'
- References 'Inflammatory Bowel Diseases' as motivation
- Studies immune pathways with clear disease application focus

Holmes (2025) - CCC SP25.txt

- Explicitly connects to Parkinson's Disease: 'Dysregulated melanogenesis expression is linked to many human neurodegenerative disorders such as Parkinson's'
- Studies neuromelanin with direct human disease relevance
- Discusses potential for 'human research into neurodegenerative symptoms'

Logan et al. (2025) - CCC FA25.txt

- Focuses on ANCE gene orthologous to human ACE gene regulating blood pressure
- Explicitly states 'we could relate our findings to the ACE gene within humans' for cardiovascular disorders
- Discusses 'cardiovascular-related pathways and potential therapeutic directions'

Meraz et al. (2023) - CCC SP23.txt

- Focuses on tyrosine kinase genes with FRK being 'involved in tumor-suppression'
- Explicitly mentions cancer relevance: 'the particular function they perform...leads to their higher expression'
- Studies tumor suppressor genes with clear disease connection

Nii et al. (2025) - CCC FA25.txt

- Explicitly states 'insight that can extend to human metabolic diseases such as obesity, fatty liver disease and iron overload disorders'
- Focuses on ACC and lipid metabolism with direct disease applications
- Connects iron-induced metabolic changes to human disease outcomes

Otala et al. (2025) - LU SP25.txt

- Explicitly states 'Fly Gut is a good model for fatty acid metabolism disorders'
- Mentions specific disorders: 'MCADD and Carnitine transporter deficiency, hypoglycemia, liver dysfunction, and muscle weakness'
- Connects gene functions to human disease mechanisms

Paderna et al. (2025) - CCC SP25.txt

- Focuses on SLC45A2 associated with 'oculocutaneous albinism type 4' in humans
- Explicitly studies albinism disease mechanism
- Connects Drosophila gene to human pigmentation disorder

Paramo-Ojeda et al. (2025) - CCC SP25.txt

- Directly states 'KRAS mutations drive many human cancers'
- Explicitly models 'KRAS-Driven Colon Cancer'
- Studies oncogene with clear cancer disease focus

Rodriguez (2025) - CCC SP25.txt

- Focuses on GALM gene associated with galactosemia
- Explicitly states 'Mutations in the protein galactose mutarotase, or GALM, are associated with the disease galactosemia'
- Studies metabolic disorder with clear disease focus

Trevino et al. (2022) - CCC SP22.txt

- Directly models Zellweger Spectrum Disorder (ZSD)
- Explicitly studies mutation of genes involved in peroxisome formation causing ZSD
- Connects Drosophila findings to human genetic disorder

Tuttle et al. (2025) - CCC SP25.txt

- Studies tryptophan-kynurenine pathway linked to aging
- Explicitly states 'kynurenine pathway has been found to have links to the degradation and aging of organisms'
- Connects gene mutations to lifespan and age-related pathology

Zlaket et al. (2023) - CCC SP23.txt

- Focuses on SCP2 gene where 'errors within this gene can cause Zellweger's Syndrome'
- Studies genetic disorder with peroxisomal dysfunction
- Connects gene function to specific human disease mechanism

Metabolic/Physiological (5 posters)

Berbouti et al. (2025) - CCC FA25.txt

- Studies trehalose catabolism pathway in Drosophila midgut
- Focus on carbohydrate metabolism and sugar processing
- Highlights connection to 'parallel sugar processing in humans' without explicit disease target

Dehuelbes et al. (2025) - LU SP5.txt

- Studies metabolic genes (CRAT, MANF, NPY) involved in energy homeostasis
- Highlights role in 'cellular energy homeostasis' and 'feeding behavior'
- Discusses therapeutic exploration 'for treating metabolic disorders in humans' but primary focus is on metabolic mechanisms

Lemus et al. (2025) - FA25 CCC.txt

- Studies amylase genes involved in starch and sucrose metabolism
- Focus on 'macromolecule breakdown' and digestive processes
- While mentions 'rectum adenocarcinoma' as potential relevance, primary focus is on digestive biology

Luera et al. (2025) - CCC FA25.txt

- Studies NPF/NPFR signaling in feeding behavior
- Focus on gut-mediated feeding behavior and energy homeostasis
- No explicit disease target; focuses on fundamental neuropeptide signaling

Pedireddi et al. (2025) - CCC SP25.txt

- Studies sugar metabolism and polysaccharide digestion
- Focus on carbohydrate breakdown processes
- While discusses human relevance, primary focus is on digestive mechanisms

Basic Biology (3 posters)

Diaz et al. (2025) - CCC SP25.txt

- Focuses on V-ATPase genes and phagosome acidification
- Studies cellular defense mechanisms in the midgut
- Makes analogy to human stomach but primary focus is on fundamental cellular processes

Ford et al. (2023) - CCC SP23.txt

- Studies Cytochrome P450 genes and insecticide resistance
- Primary focus on gene expression patterns and coexpression
- Notes Drosophila-human parallels but focus is on basic insect physiology

Sakana et al. (2025) - CCC SP25.txt

- Studies TALE homeobox genes and developmental patterning
- Focus on fundamental gene expression control and midgut development
- While mentions human MEIS genes, primary focus is on Drosophila developmental mechanisms

Posters with Mixed Category Elements

The following posters had elements that could fit into multiple categories. For these posters, the primary category assignment and reasoning for the final choice is explained.

Dehuelbes et al. (2025) - LU SP5.txt

Alternative categories considered: Could be: Disease-Motivated or Metabolic/Physiological

Assignment reasoning: Assigned to Metabolic/Physiological because primary focus is on understanding metabolic gene function (CRAT, MANF, NPY) in energy homeostasis. While it mentions therapeutic potential for metabolic disorders, the core research is mechanistic understanding of metabolic processes.

Henriquez et al. (2025) - COD WI24.txt

Alternative categories considered: Could be: Basic Biology or Disease-Motivated

Assignment reasoning: Assigned to Disease-Motivated because the introduction explicitly states the goal is understanding 'human immune disorders that reside within the gut such as Crohn's disease and related Inflammatory Bowel Diseases.' The disease application is central to the research motivation.

Holmes (2025) - CCC SP25.txt

Alternative categories considered: Could be: Basic Biology or Disease-Motivated

Assignment reasoning: Assigned to Disease-Motivated because the research explicitly connects to Parkinson's Disease and neurodegenerative disorders. The discussion states 'Dysregulated melanogenesis expression is linked to many human neurodegenerative disorders such as Parkinson's.'

Meraz et al. (2023) - CCC SP23.txt

Alternative categories considered: Could be: Basic Biology or Disease-Motivated

Assignment reasoning: Assigned to Disease-Motivated because the gene FRK (human ortholog) is explicitly described as 'involved in tumor-suppression' and the research aims to understand tumor suppressor mechanisms.

Nii et al. (2025) - CCC FA25.txt

Alternative categories considered: Could be: Metabolic/Physiological or Disease-Motivated

Assignment reasoning: Assigned to Disease-Motivated because the abstract explicitly states findings can extend to 'human metabolic diseases such as obesity, fatty liver disease and iron overload disorders.' The disease applications are central to the research justification.